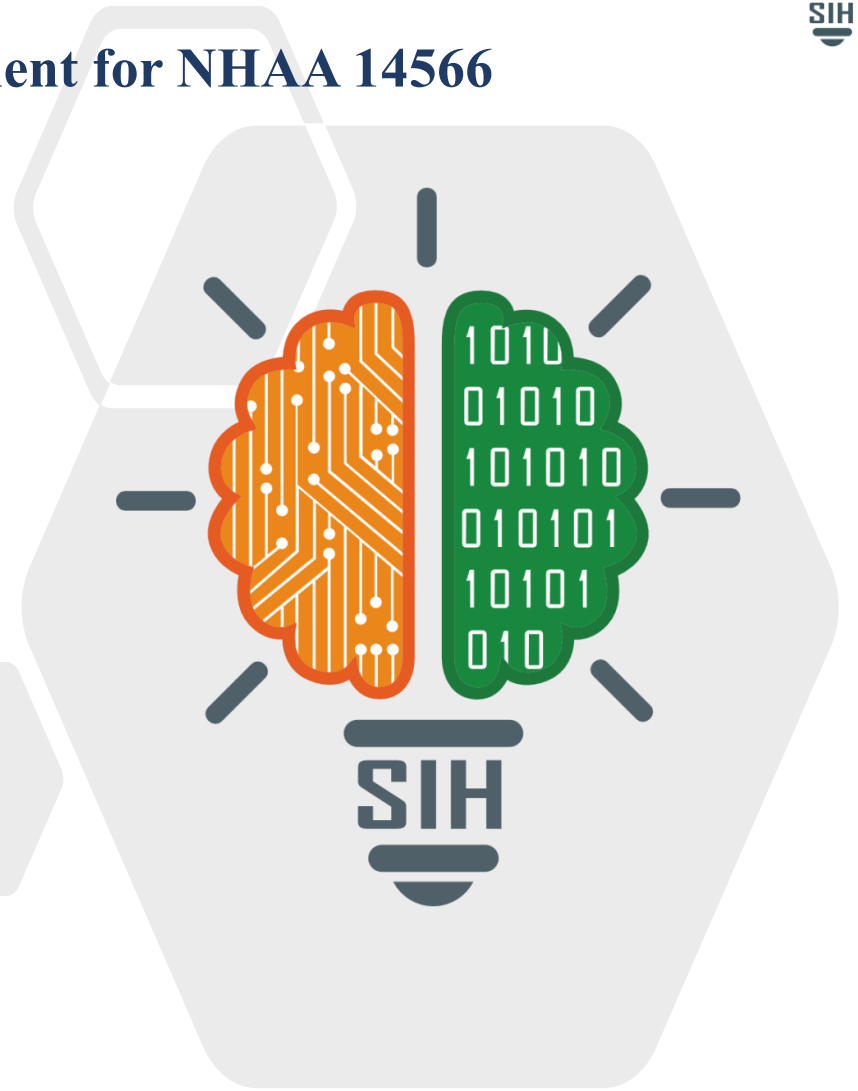
SMART INDIA HACKATHON 2026

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SAMVEDNA – AI Stress & Trauma Assessment for NHAA 14566



- **Problem Statement ID – SIH26093**
- **Problem Statement Title-** AI-Based Real-Time Stress and Trauma Assessment Module for Victims/Complainants Accessing NHAA (14566) and Integrated Portal
- **Theme-** MedTech / BioTech / HealthTech
- **PS Category-** Software
- **Team ID- 159009**
- **Team Name - Neura Forge**

▶ CLICK HERE FOR THE DEMO VIDEO

Working prototype UI · Gemma 4, to be fine-tuned by us · on-premise · every external figure cited on slide 6

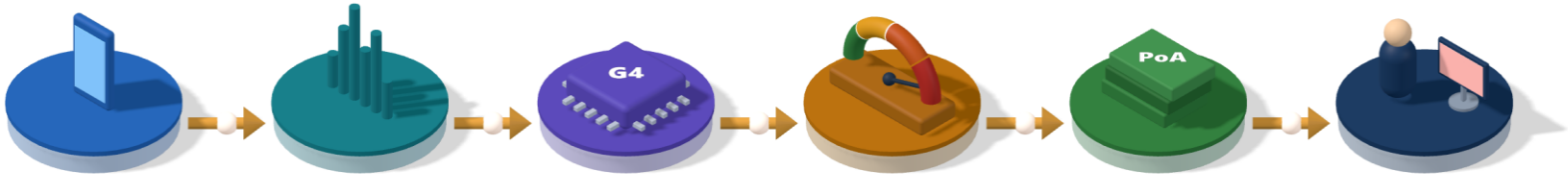
SAMVEDNA – STRESS & TRAUMA ASSESSMENT

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Detailed explanation of the proposed solution

one call, end to end, on the helpline's own server



1 CONNECT — any channel; a person asks consent

2 LISTEN — speech to text + voice cues

3 UNDERSTAND — Gemma 4 flags 7 warning signs

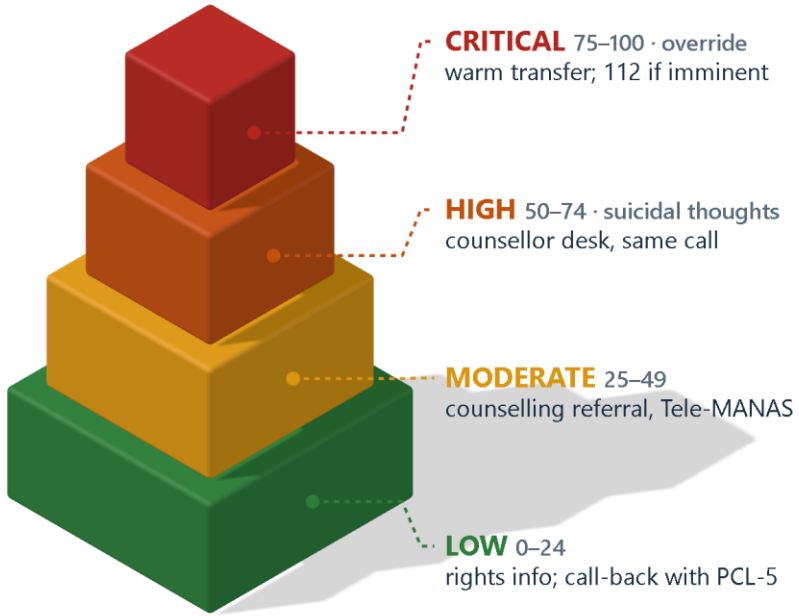
4 SCORE — SVI 0–100 + suicide override

5 ADVISE — help that cites the law

6 RESPOND — counsellor acts; all logged

Stress Vulnerability Index → action

Legal and safety actions fire on what is reported, at any band.



CRITICAL 75–100 · override warm transfer; 112 if imminent

HIGH 50–74 · suicidal thoughts counsellor desk, same call

MODERATE 25–49 counselling referral, Tele-MANAS

LOW 0–24 rights info; call-back with PCL-5

How it addresses the problem

No check at first contact [PS]	Screened in every call
Distress goes unrecorded [2]	Needs logged with the case
0.75 psychiatrists per lakh [6]	Critical answered first
Rights scattered across law [9]	Each step cites its section
85% treatment gap [5]	Tele-MANAS in the same call

Innovation and uniqueness

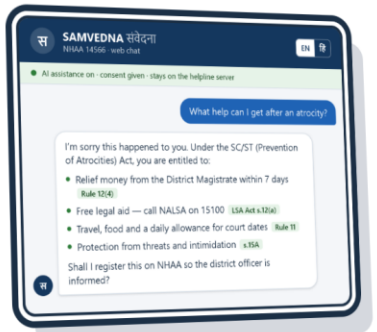
A score you can audit — screeners 40% · model 35% · voice 10%

The caller's own words — exact transcript quote, replayable

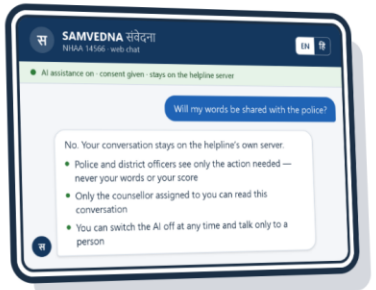
Our model, on-premise — Gemma 4, to be fine-tuned by us

Help that cites the law — s.15A · Rule 12(4) · Rule 11

PROTOTYPE UI
real chatbot answers · synthetic callers



Rights after an atrocity · **LOW**



"Will police see it?" · **PRIVACY**



Threats not to testify · **HIGH**



Suicide risk (Hindi) · **CRITICAL**

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TECHNICAL APPROACH

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Methodology and process for implementation

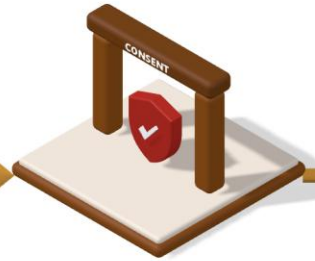
system architecture — one call flows left to right

1 CHANNELS



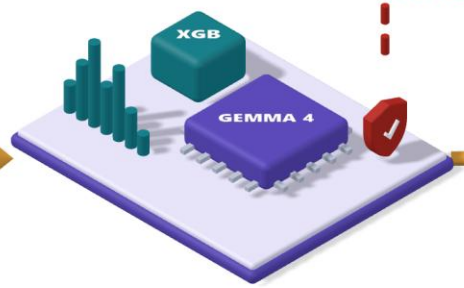
14566 IVRS · Portal
Chatbot · Mobile app

2 CONSENT GATE



consent asked by a person
declined → operator screens
under 18 → human only

3 AI ENGINE



Whisper · 8 kHz Hindi speech
eGeMAPS + XGBoost · voice **OURS**
Gemma 4 E4B + LoRA · 7 signs **OURS**
Safety net · C-SSRS + crisis phrases

4 SVI ENGINE

OVERRIDE → CRITICAL



SVI = .40S + .35L + .10V + .15C
Low · Moderate · High · Critical
PoA Act rules-as-code

5 ACT & RECORD



counsellor desk acknowledges
district alert, no clinical data
dashboards · audit ledger

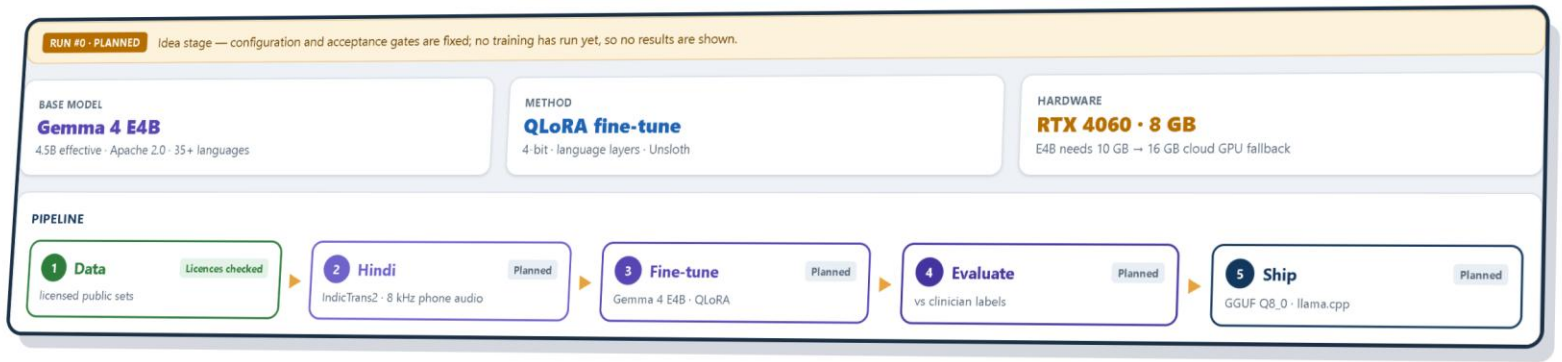
Technologies to be used

open source · on-premise

INTERFACE	React 19	TypeScript	Tailwind 4	WebSocket
SPEECH	Whisper-turbo	Silero VAD	openSMILE	IndicTrans2
AI / ML	Gemma 4 E4B	Unsloth · PEFT	llama.cpp	XGBoost
BACKEND	FastAPI	SQLite → PostgreSQL	SHA-256 audit chain	
PRIVACY	Consent · 22 languages	PII masking	RBAC	Encryption

Fine-tune dashboard

prototype UI · run #0 planned · Gemma 4 E4B QLoRA



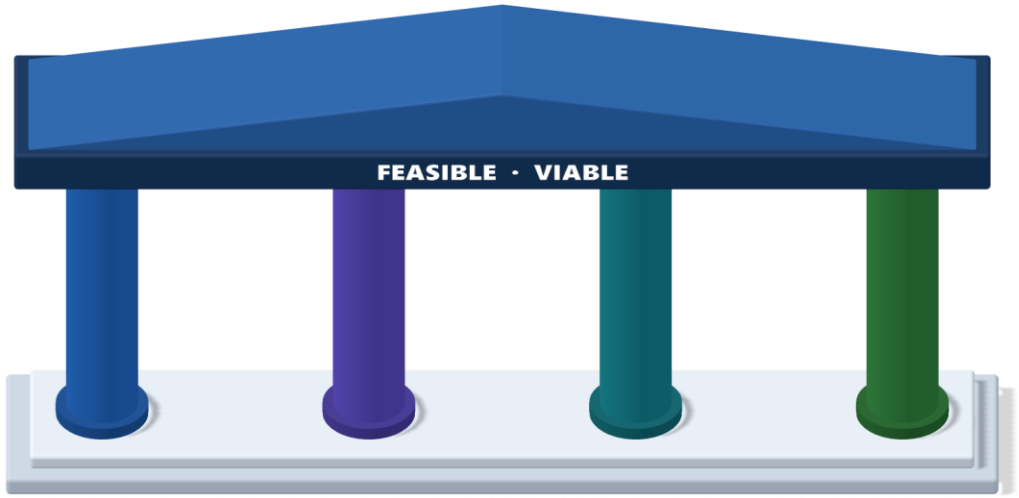
Real screenshot of our prototype: the configuration and acceptance gates are fixed; no training has run yet.

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FEASIBILITY AND VIABILITY

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Analysis of the feasibility of the idea



TECH	DATA	DEMAND	PROOF
Gemma 4 · Apache 2.0 Whisper · llama.cpp · MIT screeners free to use	Reddit C-SSRS · CC BY CREMA-D · ODbL 2 licences flagged [18]	UP: 3,40,742 calls, 54% [2] ≈ 950 calls a day at peak 1 GPU server per centre	ANUPALAN (SIH26024) 170 / 182 held-out right 0 invented Acts in 219

Plugs into what already runs: Tele-MANAS 14416 [7] · NHAA alerts [4] · 217 Special Courts [3]; adds one counsellor desk per centre

POTENTIAL CHALLENGES AND RISKS	STRATEGIES FOR OVERCOMING THEM
Voice "stress detection" is unreliable alone US field test: ~15% of lies caught [16]; hotline: 24% missed [15]	**Voice ≤ 10%, after calibration** can raise concern, never sets a suicide flag
A missed suicide risk is the worst failure PHQ-9 item 9 daily: 4% one-year attempt risk vs 0.4% [11]	**Three catches, then a person** C-SSRS for every victim, crisis phrases, model; never left alone
Phone Hindi is hard to transcribe even an Indic Whisper: 26.8% word error rate [13]	**Never rely on one transcript** Indic-tuned Whisper; low confidence down-weighted; keypad, chat
No Indian clinical distress data no Hindi depression dataset among 108 surveyed [17]	**Licensed now, consented later** public sets now; pilot adds opt-in, ethics-approved calls
Staff may over-trust the AI WHO warns of "automation bias" [20]	**AI recommends, a counsellor decides** every flag shows its quote; ICMR human-in-the-loop
Caste and mental-health data are high-risk DPDP s.6 consent; MHCA s.23 confidentiality [19]	**Clinical data stays on the console** on-premise; police see actions, never quotes or scores
Our training GPU has 8 GB Gemma 4 E4B needs 10 GB to fine-tune [12]	**Lean training, cloud fallback** 4-bit LoRA on the RTX 4060; free 16 GB cloud GPU if needed

Roadmap from this idea to national scale

0 · IDEA
this deck, Sep 2026

1 · PROTOTYPE
voice + text pipeline, console, synthetic demo

2 · PILOT
one State, ethics-approved, with clinicians

3 · SCALE
all States, Tele-MANAS link, dashboards

4 · SUSTAIN
retrain on consented data, yearly audit



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IMPACT AND BENEFITS

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The scale of need
sources on slide 6

65,664 atrocity cases against SCs and STs, 2024 [1]

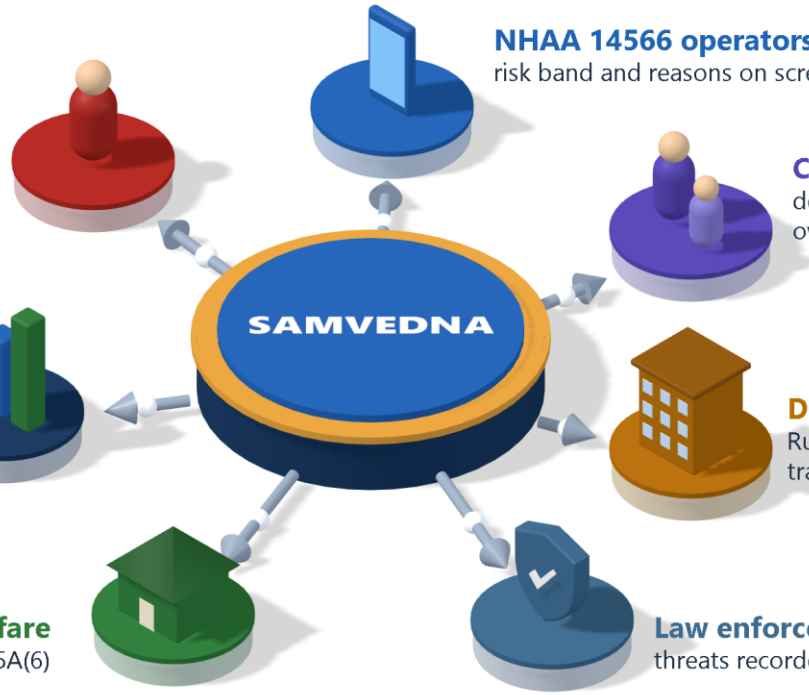
95.2% of SC cases in court pending trial, 2024 [1]

6,34,066 calls to 14566 by Jul 2025 [2]

85% treatment gap, depression and anxiety [5]

1,70,746 suicides in India, 2024 [8]

Potential impact on the target audience
victims + every stakeholder the PS names



Victims & complainants
heard, screened, told their rights in one call

NHAA 14566 operators
risk band and reasons on screen

Counsellors
desk per centre; Tele-MANAS overflow

DoSJE, States & UTs
distress and response, counts only

District administrations
Rule 12(4) relief, Rule 11 travel flagged

Rehabilitation & welfare
Special Court application, s.15A(6)

Law enforcement
threats recorded under s.15A(9)

Benefits of the solution

SOCIAL
dignity at first contact, s.15A(2)

HEALTH
designed to catch suicide risk in the call

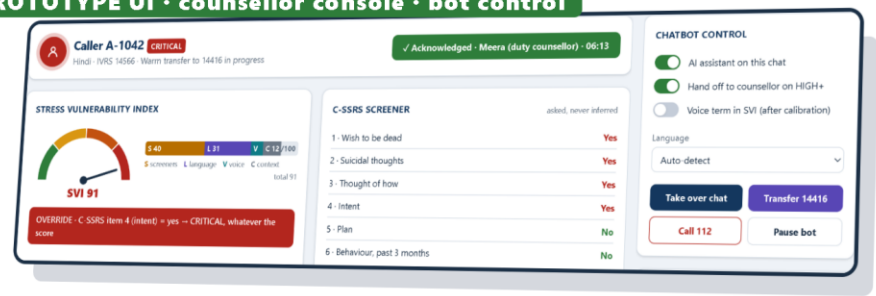
JUSTICE
aims to keep witnesses supported

ECONOMIC
no per-call or cloud fees

GOVERNANCE
district distress data, audit trail

ENVIRONMENTAL
paperless, one server per centre

PROTOTYPE UI · counsellor console · bot control



PILOT TARGETS
to be measured, not results

≥ 0.95
sensitivity for High and Critical; every miss reviewed

$\kappa_W \geq 0.6$
band agreement with a blinded clinician

< 60 s
to a full SVI; overrides fire at once

WER
on consented 14566 audio, per language

0
audio, quotes or scores leave the premises

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RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Details / links of the reference and research work

every external figure in this deck comes from these sources; a second reviewer opened each one

problem & policy · law · clinical · AI & speech · privacy & ethics

1. **NCRB — Crime in India 2024, Vol. II (2026)**
65,664 atrocity cases against SCs and STs in 2024; 95.2% of SC cases in court pending trial; SC conviction rate 33.9%.
ncrb.gov.in — 2CrimeinIndia2024-VolumeII.pdf, Tables 7A, 7C

2. **Rajya Sabha Unstarred Q. 2063, MoSJE (6 Aug 2025)**
6,34,066 calls to NHAA as on 31 Jul 2025; 7,661 complaints, filed only as FIR, charge sheet or relief; UP 3,40,742 calls.
sansad.in — AU2063, Annexures 1 and 2

3. **MoSJE — Annual Report 2025-26, section 3.9**
NHAA registered 8,257 grievances since Dec 2021; 217 Exclusive Special Courts; 181 Special Police Stations.
socialjustice.gov.in — Annual Report 2025-26

4. **PIB — NHAA 14566 launched (13 Dec 2021)**
Round the clock, in Hindi, English and regional languages; IVR, web portal and app; automatic alerts to States.
pib.gov.in — PRID 1780979

5. **National Mental Health Survey 2015-16 (Murthy 2017)**
10.6% of adults have a current mental disorder; the treatment gap for depression and anxiety is 85%.
pmc.ncbi.nlm.nih.gov — PMC5419008

6. **PIB backgrounder — mental health (9 Nov 2025)**
0.75 psychiatrists per lakh people in India, against WHO's recommended 3 (citing Garg et al. 2019).
pib.gov.in — PRID 2188003

7. **PIB factsheet on healthcare (12 Aug 2026)**
Tele-MANAS 14416: 43.64 lakh calls, 20 languages, 53 cells, as on 11 Aug 2026.
pib.gov.in — Factsheet 150834

8. **NCRB — ADSI 2024, chapter 2: suicides**
1,70,746 suicides in India in 2024, a rate of 12.2 per lakh population.
NCRB PDF via data.opencity.in mirror

9. **SC/ST (PoA) Act 1989, amended by Act 1 of 2016**
'Victim' includes an SC/ST person harmed mentally, psychologically or emotionally by an offence, and relatives; s.15A: victims' rights.
prsindia.org — Gazette of India, 1 Jan 2016

10. **PoA Rules 1995, amended by G.S.R. 424(E), 2016**
Rule 12(4): relief within 7 days (Annexure-I: rape ₹5 lakh, murder ₹8.25 lakh); Rule 11: travel, daily allowance, diet.
poa.apcfss.in — G.S.R. 424(E) · tribal.nic.in — 1995 Rules

11. **PHQ-2/9, GAD-2/7, C-SSRS (Posner 2011), PCL-5**
PHQ-2/GAD-2 positive at ≥ 3; PHQ-9 item 9 "nearly every day" → 4% one-year attempt risk vs 0.4% (Simon 2013).
Psychiatr Serv 2013 · phqscreeners.com · cssrs.columbia.edu · ptsd.va.gov

12. **Gemma 4 model card (2026); Unsloth guide**
E4B: 4.5B effective parameters, Apache 2.0, 35+ languages (140+ pre-trained), audio input; E4B fine-tuning needs 10 GB.
ai.google.dev/gemma · unsloth.ai/docs/models/gemma-4

13. **Vistaar (Interspeech 2023); Whisper (ICML 2023)**
On telephone Hindi, even an Indic Whisper fine-tune has a 26.8% word error rate; original Whisper trained on 680,000 hours.
arXiv 2305.15386 · proceedings.mlr.press/v202

14. **eGeMAPS (Eyben et al., IEEE TAC 2016)**
A standard set of 88 voice parameters — pitch, jitter, shimmer, loudness, pauses — chosen for their link to emotion.
sail.usc.edu — eyben-preprinttaffc-2015.pdf

15. **JMIR 2025: hotline acoustics; speech meta-analysis**
The hotline study missed 24% of high-risk audio segments; 105 studies: speech analysis is "complementary", not standalone.
doi 10.2196/67772 · doi 10.2196/67802

16. **US NIJ — voice stress analysis field test (2008)**
Voice-stress analysers caught ~15% of arrestees' lies about recent drug use — near chance.
nij.ojp.gov — NIJ Journal 259

17. **Bucur et al. — mental-disorder dataset survey (2025)**
108 datasets, 25 languages, 23 freely downloadable; none for Hindi depression.
arXiv 2505.15556

18. **Training datasets and their licences**
Reddit C-SSRS (500 users, CC BY 4.0); RAVDESS (CC BY-NC-SA); CREMA-D (ODbL); Rasa (CC BY 4.0); Dreaddit; DAIC-WOZ, UMD gated.
zenodo 2667859 · zenodo 1188976 · github CREMA-D, AI4Bharat/Rasa

19. **DPDP Act 2023 & Rules 2025; MHCA 2017; POCSO 2012**
Consent free, specific, informed (s.6); emergency s.7(f); children s.9; 72-hour breach report (Rule 7, from 2027); MHCA s.23; POCSO s.19.
egazette.gov.in · meity.gov.in — G.S.R. 846(E)

20. **WHO LMM guidance (18 Jan 2024); ICMR AI ethics (2023)**
WHO's guidance (over 40 recommendations) warns LMMs can encourage "automation bias"; ICMR calls for a human in the loop.
who.int · icmr.gov.in — AI guidelines 2023

21. **Legal Services Authorities Act 1987; NALSA**
An SC/ST member who has to file or defend a case gets free legal services, whatever their income (s.12(a); prima facie case, s.13(1)); helpline 15100.
nalsa.gov.in — LSA Act 1987 · helpline 15100

Checked against the sources on 26–27 Sep 2026. ANUPALAN figures (slide 4) are our own SIH26024 evaluation. Slide 2 caller data is synthetic.

Team Neura Forge · SIH26093